Programming Fundamentals II: Lab 04

Due Date: Tuesday October 01, 2024 at Midnight

Part 1. Fare Enumeration (10 Marks)

For this lab, we are going to simulate fares for selling train tickets. Depending on what type of fare you qualify for, you will pay a different rate. The fare you pay is determined by your age. We have 4 types of fares, as follows:

Fare Type	Rate	Min Age	Max Age
Infant	$4.50	0	4
Youth	$9.40	5	17
Adult	$15.00	18	64
Senior	$10.65	65	150

a. **(4 Marks)** Create an Enum named Fare to handle the four Fare types listed above.

b. **(3 Marks)** Give your Enum a static method named "determineFare". This method should accept somebody's age as input, it will then return the Fare type that that person should pay. An age over 150 or under 0 is considered to be *illegal.*

c. **(3 Marks)** Download the Lab04.java file provided. In this file, we have a lists of people and their ages. For each person, determine their fare type and how much that person will need to pay. Use a StringBuilder to assemble the final passenger list, which will list the passengers and their fare types and rates. Display the results in this format:

```
Fares:
   Alice: $15.0, ADULT
   Bob: $10.65, SENIOR
   Charlie: $4.5, INFANT
   David: $9.4, YOUTH
   Eden: $9.4, YOUTH
   Frank: $15.0, ADULT
   George: $10.65, SENIOR
```

Add the **date and your name and student number** to the top of the java files. Please submit your .java files to Nexus before the end of the day on the due date. **Late submissions will be not be accepted.**